Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Assad, Christopher Ph.D.		
Project Title:	Fault detection signal analysis and system modeling for AEMC		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	12/09/2002	**End Date:**	09/26/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Daud, Taher (Jet Propulsion Laboratory)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We will apply a novel signal analysis method to enhance fault detection for Advanced Environmental Monitoring and Control (AEMC) systems. Failures of critical components during operation of Advanced Life Support (ALS) subsystems should produce transient and/or long-term changes in their associated sensor feedback. These changes need to be detected and characterized as the first step in maintaining system safety and efficient operation. Recently Prof. Gautam Biswas at Vanderbilt has been funded by NASA to refine an AEMC control framework that includes a fault detection system to monitor ALS subsystem activity, as well as an active model component. We are developing general-purpose tools to apply a new signal analysis method to enhance the fault detection system by analyzing spectral and temporal characteristics of sensor data streams. The new methods have potential advantages to enable better spectral analysis of data generated by nonlinear, non-stationary processes. We will compare the new methods to state-of-the-art methods that use Fourier and wavelet analyses for fault detection. We plan to evaluate the results on available data and models of the ALS Advanced Water Recovery System (AWRS) at Johnson Space Center. We will also survey adaptive nonlinear		

	system identification methods for potential application to active modeling of the AWRS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goals of this task are to develop methods to improve fault detection and systems modeling for AEMC systems. Earth benefits may include a wide range of applications requiring advanced monitoring methods, such as water processing and medical monitoring.
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)